Pitch Control Moment Calculation Handoff

Static trim and dynamic maneuver requirements using the optimized trajectory

This procedure defines the common analysis that must be completed for all five candidate center of gravity locations. The objective is to calculate the pitch-control moment required for static trim and for dynamic tracking of the optimized trajectory. These moment histories will become the standard inputs for preliminary control-surface sizing and later deflection calculations.

Required deliverables

Deliverable	Required content
Excel calculation workbook	Imported standardized trajectory data, visible formulas, constants, units, and calculations for all five CG values.
Time history data	Static trim and dynamic maneuver moment requirements in both N m and coefficient form at every trajectory point.
Summary table	Positive, negative, and absolute maximum requirements with the associated time, Mach, altitude, angle of attack, and dynamic pressure.
Plots	Required moment versus time and required moment coefficient versus Mach, with all five CG cases clearly identified.
Verification record	Confirmed values and sources for Sref, Cbar, Iyy, CG coordinates, moment reference, units, and sign conventions.

Pre analysis verification gate

Do not begin the calculations until the reference values have been checked against the specific trajectory and aerodynamic dataset being used. Confirm the values with the appropriate aerodynamics, CAD, structures, and mass properties sub teams. Record who or what source confirmed each value in the Excel workbook.

Check	Working value	Required confirmation
Reference area Sref	Read from Aero coeffs.csv; current file contains 0.02556109 m^2	Confirm with the aerodynamic dataset owner.
Reference length Cbar	0.18038 m	Confirm with aerodynamics and CAD. Update the MATLAB script before generating the common output.
Pitch inertia Iyy	15.536 kg m^2	Confirm with structures or mass properties and verify that it is about the pitch axis through the CG.
Moment reference	Current script assumes the sharp nose at xref = 0 m	Confirm the aerodynamic moment reference and longitudinal coordinate direction.
CG locations	1.08, 1.12, 1.16, 1.20, and 1.24 m	Confirm that all values use the same origin and coordinate convention as the aerodynamic data.

Completion gate: all five checks above must be documented before the results are accepted.

Standard data pipeline

Abhay recommends running the MATLAB interpolation once, freezing the resulting CSV as the standard input, and then performing the required moment calculations in Excel. All five CG cases must use the same interpolated trajectory and clean aerodynamic coefficient history.

Required source files

- Optimized trajectory: Designs/Lifting Body V2/bulge_results_collocation.csv
- Aerodynamic database: Designs/Lifting Body V2/Aero coeffs.csv
- Trajectory processing script: GNC Technical Documents/InterpolateTraj.m
- Aerodynamic lookup function: GNC Technical Documents/interpolateAeroTrajectory.m
- Standard output: GNC Technical Documents/pitch_maneuver_history_with_aero.csv

Interpolation sequence

1. Read and time-sort the optimized trajectory.
2. Create a common time grid with spacing no greater than 0.25 s.
3. Smooth angle of attack and flight-path angle with cubic smoothing splines.
4. Calculate pitch attitude, pitch rate, and pitch acceleration from the spline histories.
5. Interpolate Mach, speed, dynamic pressure, and altitude with shape-preserving cubic interpolation.
6. Linearly interpolate the clean aerodynamic coefficients at each paired Mach, altitude, and angle-of-attack trajectory state. Extrapolation is prohibited.
7. Translate the clean nose-referenced pitching moment to each candidate CG and write one clean-Cm column per CG to the common output file.

CG input warning

The current MATLAB script defines xcgCandidates as nondimensional xCG divided by vehicle length and then multiplies by vehicle length. The approved candidates above are dimensional values in metres. Do not paste the metre values into the nondimensional array without removing the additional multiplication by vehicle length. The common output should be generated only after this interpretation is corrected and independently checked.

What remains common and what changes

Common for all CG cases	Changes with CG
Time, Mach, altitude, speed, dynamic pressure, angle of attack, flight-path angle, pitch attitude, pitch rate, and pitch acceleration	Clean pitching moment about the selected CG
Clean CN, CA, CL, CD, pressure drag, and skin-friction drag	Static trim moment requirement
Optimized baseline trajectory	Dynamic maneuver moment requirement
Interpolation methods and aerodynamic database	Later control-surface deflection and trim-drag histories

The baseline trajectory must not be re-interpolated separately for each CG during this initial comparison. A CG-specific trajectory is considered only after control-surface deflections and their lift and drag increments are included and the trajectory is reoptimized.

Excel calculation procedure

Import the standardized pitch_maneuver_history_with_aero.csv into Excel as a table. Preserve the original interpolated columns. Add calculation columns to the right and use named cells or named ranges for Sref, Cbar, and Iyy so that the constants are visible and auditable.

Candidate center of gravity locations

The calculation must be completed for all of the following dimensional CG locations:

Case	xCG m
CG 1	1.08
CG 2	1.12
CG 3	1.16
CG 4	1.20
CG 5	1.24

Translate the clean moment to the CG

For the common aft-positive longitudinal convention used by the current script:

$$C_{m,CG}(t) = C_{m,nose}(t) + C_N(t)\frac{x_{CG} - x_{ref}}{\bar{c}}$$

Use the corresponding clean-Cm output column for each CG case. Before proceeding, verify the sign convention by confirming that the CG shift produces the expected change in static stability.

Calculate the clean aerodynamic moment

$$M_{clean}(t) = q_\infty(t) S_{ref} \bar{c} C_{m,CG}(t)$$

Excel structured-reference example:

```
=[@DynamicPressure_Pa]*Sref*Cbar*[@CmClean]
```

Static trim requirement

Static trim requires zero net pitching moment. This calculation demonstrates compliance with the pitch-trim requirement and does not use Iyy.

$$M_{trim,required}(t) = -M_{clean}(t)$$

$$\Delta C_{m,trim}(t) = -C_{m,CG}(t)$$

Excel examples:

```
M_trim_required_Nm: =-[@M_clean_Nm]
DeltaCm_trim: =-[@CmClean]
```

Dynamic maneuver requirement

The dynamic calculation determines the control moment required to follow the pitch acceleration in the optimized trajectory. It supplements the static trim calculation; it does not replace it.

$$M_{maneuver,required}(t) = I_{yy}\ddot{\theta}(t) - M_{clean}(t)$$

$$\Delta C_{m,maneuver}(t) = \frac{I_{yy}\ddot{\theta}(t)}{q_{\infty}(t)S_{ref}\bar{c}} - C_{m,CG}(t)$$

Excel examples:

```
M_maneuver_required_Nm: =Iyy*[@PitchAccel_rad_s2]-[@M_clean_Nm]
DeltaCm_maneuver: =[@M_maneuver_required_Nm]/([@DynamicPressure_Pa]*Sref*Cbar)
```

Required result columns

Column	Description	Unit
Time_s	Standard interpolated time	s
Mach	Trajectory Mach number	-
Altitude_m	Trajectory altitude	m
Alpha_deg	Trajectory angle of attack	deg
DynamicPressure_Pa	Trajectory dynamic pressure	Pa
PitchAccel_rad_s2	Commanded pitch angular acceleration	rad/s^2
CmClean	Clean pitching-moment coefficient at the selected CG	-
M_clean_Nm	Clean aerodynamic pitching moment	N m
M_trim_required_Nm	Control moment required for static trim	N m
DeltaCm_trim	Required static control moment coefficient	-
M_maneuver_required_Nm	Control moment required for trajectory tracking	N m
DeltaCm_maneuver	Required dynamic control moment coefficient	-

Interpretation: a positive required moment demands control authority in the positive pitch direction; a negative value demands the opposite direction. Label those directions as nose-up or nose-down only after the project sign convention is confirmed.

Summary and quality checks

Required summary for each CG case

Reported quantity	Associated flight condition
Maximum positive static trim moment	Time, Mach, altitude, AoA, and dynamic pressure
Maximum negative static trim moment	Time, Mach, altitude, AoA, and dynamic pressure
Maximum absolute static trim moment	Time, Mach, altitude, AoA, and dynamic pressure
Maximum positive dynamic maneuver moment	Time, Mach, altitude, AoA, and dynamic pressure
Maximum negative dynamic maneuver moment	Time, Mach, altitude, AoA, and dynamic pressure
Maximum absolute dynamic maneuver moment	Time, Mach, altitude, AoA, and dynamic pressure

Required plots

- Static trim moment versus time with all five CG cases shown.
- Dynamic maneuver moment versus time with all five CG cases shown.
- Required static moment coefficient versus Mach.
- Required dynamic moment coefficient versus Mach.
- Pitch acceleration versus time, inspected for spline-derived spikes or oscillations.

Final quality checklist

[ ] Sref, Cbar, and Iyy were independently confirmed for the specific dataset before analysis.

[ ] All five CG locations were evaluated using the same standardized interpolated trajectory.

[ ] Dynamic pressure remained in pascals throughout the Excel calculations.

[ ] Pitch acceleration remained in radians per second squared.

[ ] The moment reference, CG origin, and positive-moment convention were recorded.

[ ] Static trim and dynamic maneuver requirements were calculated and reported separately.

[ ] Maximum positive, negative, and absolute values were checked directly against the time histories.

[ ] No aerodynamic extrapolation occurred outside the database bounds.

[ ] Any spikes in pitch acceleration or dynamic control moment were investigated before reporting maxima.

Next design step

These results are moment requirements, not control-surface deflections. They will be used to select an initial surface length and then to solve for the required deflection using a control-surface aerodynamic model or CFD database:

$$\Delta C_{m,control}(M, h, \alpha, \delta, L_{CS}) = \Delta C_{m,required}(t)$$

The static trim requirement and the dynamic maneuver requirement must each be checked against the available surface authority, allowable deflection, actuator rate, and actuator torque. After deflection-dependent lift and drag are available, the trajectory should be reoptimized for the shortlisted CG and surface configurations.